7.2.

$$\varepsilon = \frac{2\pi c\hbar}{L}(n_x^2+n_y^2+n_z^2) = \frac{2\pi c\hbar}{V^{\frac{1}{3}}}(n_x^2+n_y^2+n_z^2)$$

$$p = -\sum_i a_i \frac{\partial \varepsilon}{\partial V} = -\sum_i a_i \cdot 2\pi c\hbar(n_x^2+n_y^2+n_z^2)\cdot(-\tfrac{1}{3})V^{-\frac{1}{3}}V^{-1}.$$

$$= -\sum_i a_i \varepsilon \cdot -\tfrac{1}{3}V^{-1}.$$

$$= \tfrac{1}{3}\sum_i a_i \varepsilon_i V^{-1}.$$

$$= \tfrac{1}{3}UV^{-1}.$$

ε_i

$U = \sum_i a_i \varepsilon_i$ ~

7.7. 考玻尔兹曼熵. $S = k\ln\Omega$

阿伏伽德罗常数. ↓ k

Ω ← 微观量子态数.

$$\Omega = C_N^n = \frac{N!}{n!(N-n)!}$$

$$S = k\ln\Omega = k\ln\frac{N!}{n!(N-n)!} = k(\ln N! - \ln n! - \ln(N-n)!)$$

斯特林公式. $\ln m! = m(\ln m - 1)$. $(m \gg 1)$.

$$S = k[N(\ln N - 1) - n(\ln n - 1) - (N-n)(\ln(N-n) - 1)]$$

$$= k(N\ln N - n\ln n - (N-n)\ln(N-n) - N + n + N - n]$$

$$= k(N\ln N - n\ln n - (N-n)\ln(N-n))$$

$$F = U - TS = nW - TS. = nW - Tk(N\ln N - n\ln n - (N-n)\ln(N-n)).$$

$$dF = +Tk\left[\ln n + 1 - \ln(N-n) - 1\right] + W.$$

$$= Tk\left(\ln n - \ln(N-n)\right) + W = 0$$

$$\ln\frac{n}{N-n} = -\frac{W}{Tk} \qquad \ln\frac{N-n}{n} = \frac{W}{Tk} = \ln\left(\frac{N}{n}-1\right)$$

$$n = N e^{-\frac{W}{Tk}}$$

核心：$S = k\ln\Omega$ $\qquad U = \sum_i a_i \varepsilon_i$ $\qquad Z_1 = \sum_i \omega_i e^{-\beta\varepsilon_i}$

$$= -N\frac{\partial}{\partial\beta}\ln Z_1$$

$F = U - TS$.

$$\Omega = C_N^n = \frac{N!}{n!(N-n)!}$$

$$S = k\ln\Omega = k\left(N\ln N - n\ln n - (N-n)\ln(N-n)\right)$$

$$p = \frac{NkT}{V} \qquad \sigma = \frac{n}{S} = \frac{n}{L^2}$$

$$= \frac{NkT}{L^3}$$

$$\Rightarrow L = \left(\frac{NkT}{p}\right)^{\frac{1}{3}} \qquad \sigma = n\left(\frac{p}{NkT}\right)^{\frac{2}{3}}$$



7.10.

$$N\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \int e^{-\frac{m}{2kT}(p_x^2+p_y^2+p_z^2)}\, dp_x\, dp_y\, dp_z.$$

$p = mv$

$$N\left(\frac{1}{2\pi kTm}\right)^{\frac{3}{2}} e^{-\frac{1}{2kTm}(p_x^2+p_y^2+p_z^2)}\, dp_x\, dp_y\, dp_z. \qquad E = \frac{3}{2}kT$$

$$\int \frac{A}{h^2} e^{-\alpha-\frac{1}{2kTm}(p_x^2+p_y^2)}\, dp_x\, dp_y = N$$

$$\frac{N}{V}\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m}{2kT}(v_x^2+v_y^2+v_z^2)}\, dv_x\, dv_y\, dv_z.$$

$$4\pi \frac{N}{V}\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m}{2kT}v^2} v^2\, dv$$

$$\int \frac{S}{h^2} e^{-\alpha-\frac{1}{2kTm}(p_x^2+p_y^2)}\, dp_x\, dp_y = N$$

$$\int \frac{L}{h} e^{-\alpha-\frac{1}{2mkT}p^2}\, dp = N. \Longrightarrow \alpha.$$

7.17.

$$Z_1 = \sum_i \omega_i e^{-\beta\varepsilon_i} \qquad Z_1 = Z_1^t \cdot Z_1^v \cdot Z_1^r$$

$$= \int \frac{1}{h^3} e^{-\beta\varepsilon} dx dy dz dp_x dp_y dp_z \cdot$$

V^N

正则分布

$$Z = \frac{1}{N! h^N} \int e^{-\beta \sum_{i=1}^{N} \varepsilon_i} dp_1 \cdot dp_{3N} dq_1 \cdots dq_{3N}$$

$$= \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3}{2}N}$$

理想气体

$$P = \frac{N}{\beta} \frac{\partial}{\partial V} \ln Z_1 = NkT$$

$$U = -N \frac{\partial}{\partial \beta} \ln Z_1$$

平动: $Z_1^t = V \left(\frac{2\pi m}{h^2 \beta}\right)^{\frac{3}{2}}$

$$U = -\frac{\partial}{\partial \beta} (\ln Z)$$

$$P = \frac{1}{\beta} \frac{\partial}{\partial V} \ln Z$$

$$S = k \left(\ln Z - \beta \frac{\partial}{\partial \beta} \ln Z\right)$$

$$F = -kT \ln Z$$

化学势 $u = N_A \left(\frac{\partial F}{\partial N}\right)_{T,V} = -RT \frac{\partial}{\partial N} \ln Z$.